Scalable Portals for Distant Object Manipulation in Virtual Reality

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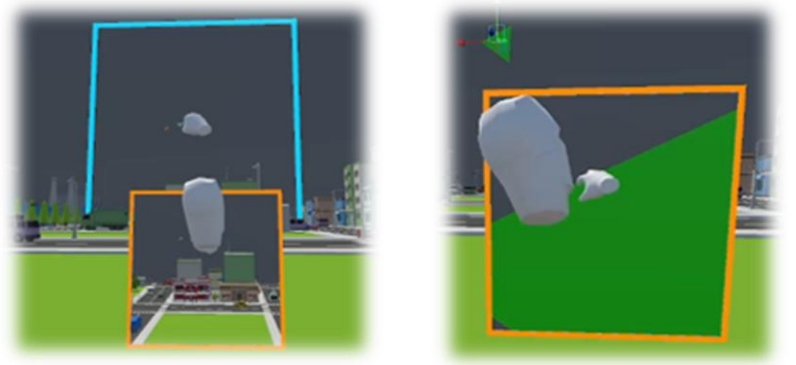
BACKGROUND & MOTIVATION

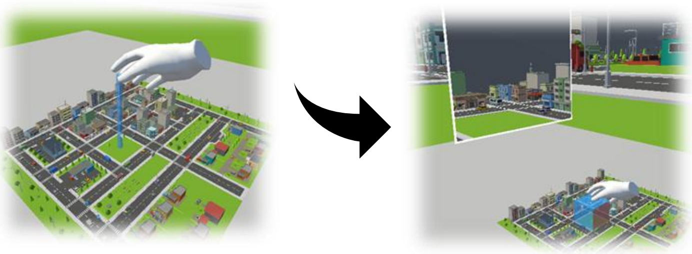
- **Problem Statement**
  - Portal interaction [1] allows users to manipulate distant objects while remaining in place [2], but its effectiveness is highly affected by the size of objects [3] beyond the portal.
  - Small targets reduce precision inside the portal (a), while large targets exceed comfortable manipulation space (b).
- **Proposed Method**
  - We propose scalable portal designs that preserve portal-based interaction while allowing users to adapt the interaction scale to support effective interaction in the current task.



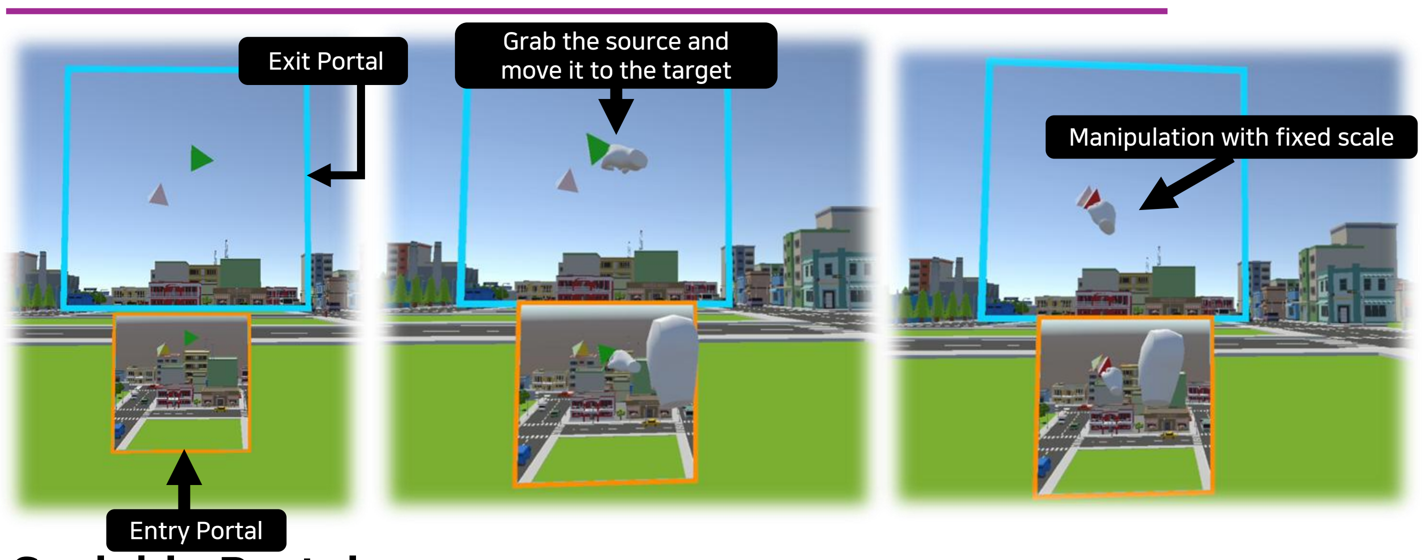
a) Small Targets b) Large Targets

SCALE TECHNIQUE OVERVIEW

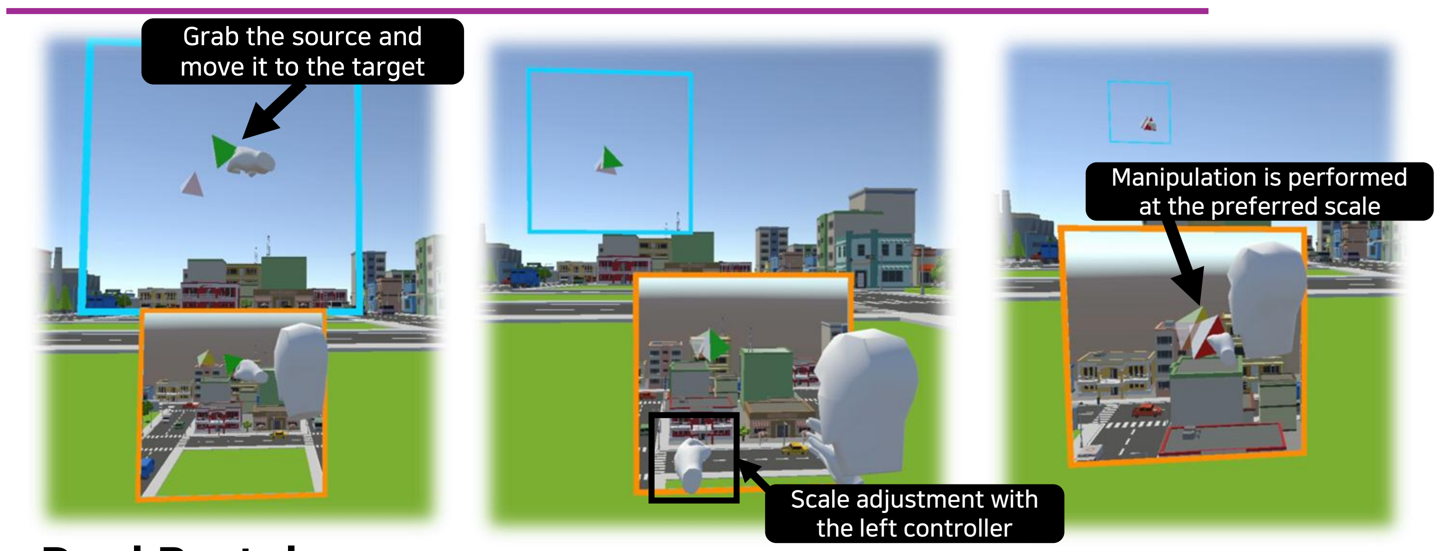
- A WiM [4] and cube-based selection defines the portal volume, which is then instantiated at a reachable position in front of the user.



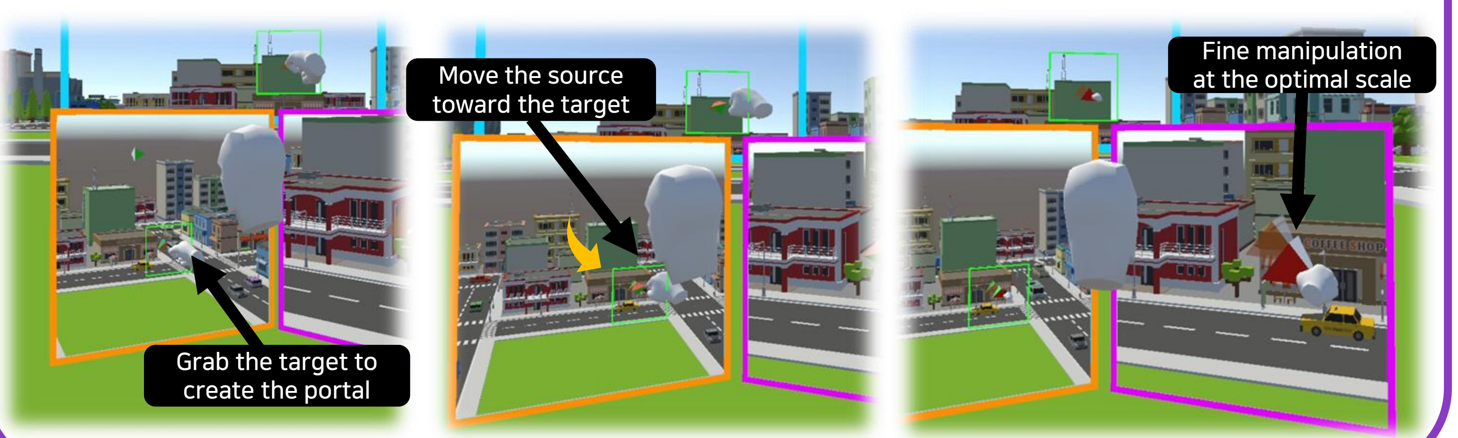
Non-Scalable Portal: Fixed size that does not change during interaction



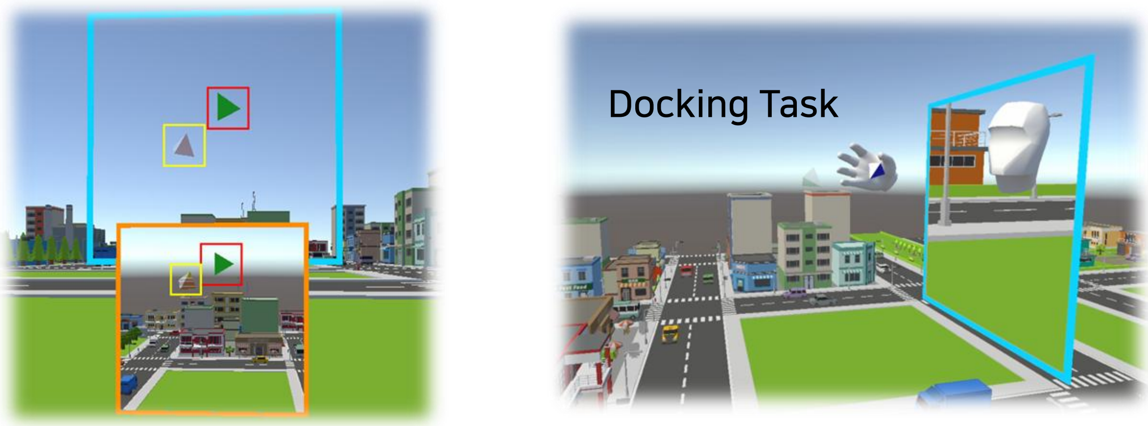
Scalable Portal: Lets the user adjust its scale for easier selection and manipulation



Dual Portal: A two-portal design that separates selection and manipulation

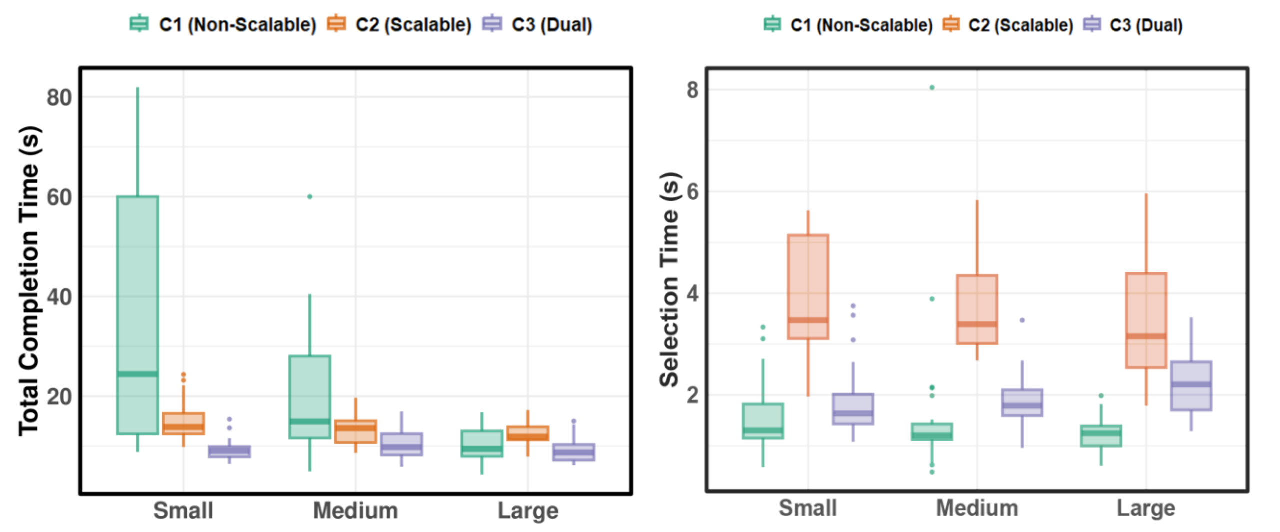


EXPERIMENT



Experimental Setup
- Pilot study with six VR-experienced participants.
- 3D tetrahedron docking task with each portal technique.
- Compared three techniques (Non-Scalable, Scalable, Dual) and three target sizes(Large, Medium, Small).
- Measured total time, selection time, and manipulation time.

RESULTS



Results and Discussion
- Scalable Portal and Dual-Portal reduced total completion time compared to Non-Scalable Portal.
- The improvement was more noticeable for smaller targets.
- Dual-Portal showed the most consistent performance across different target sizes.
- Although scalable techniques increased selection time slightly, they enabled faster overall manipulation.

CONCLUSION AND FUTURE WORK

- Scalable portals made distant object manipulation easier and more efficient.
- Both Scalable Portal and Dual-Portal performed better than Non-Scalable Portal.
- Future work will explore accuracy, workload, and improved scale-control strategies.

REFERENCES

[1] D. Han, D. Kim, and I. Cho, "PORTAL: Portal widget for remote target acquisition and control in immersive virtual environments," VRST, 2022.
[2] D. Ablett et al., "Point&Portal: A new action-at-a-distance technique for virtual reality," ISMAR, 2023.
[3] I. Cho, J. Li, and Z. Wartell, "Multi-scale 7DoF view adjustment," IEEE TVCG, 24(3):1331-1344, 2017.
[4] R. Stoakley, M. Conway, and R. Pausch, "Virtual reality on a WIM: interactive worlds in miniature," CHI, 1995.

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